

Decentralized File Storage using Blockchain: Implementation and Proof- of-Work Analysis

Guided By
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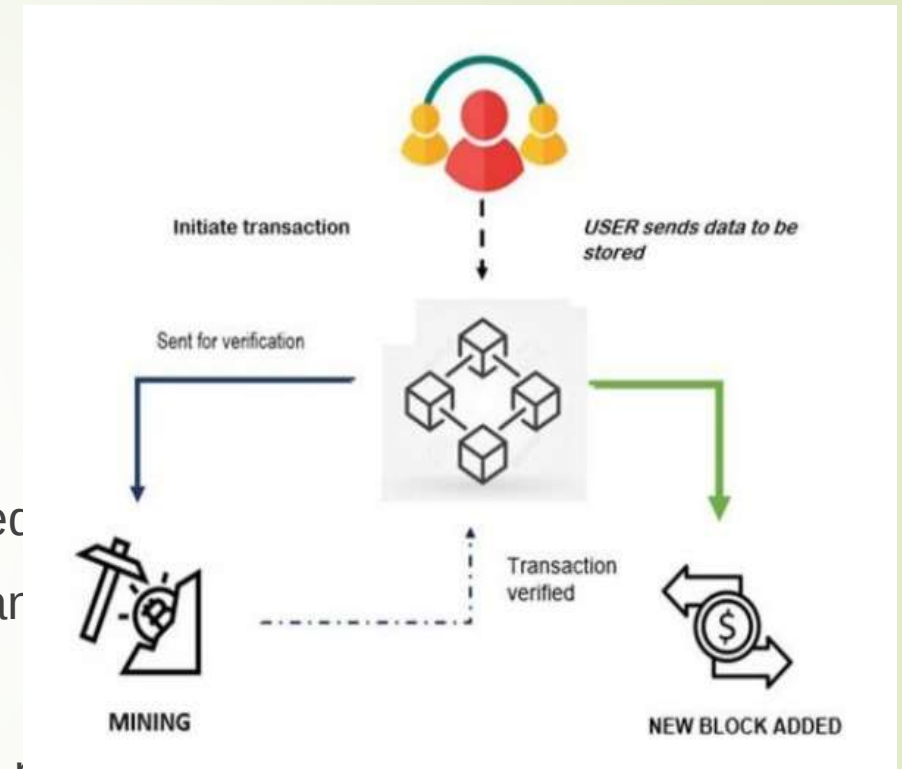
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ABSTRACT

This project delves into the development and implementation of a web-based decentralized file storage application leveraging blockchain technology for storing Medical Records.

The project's primary goal was to design a robust system enabling users to upload multiple files, each securely stored in a block within the blockchain, guaranteeing data integrity and immutability. This holds particular significance in the healthcare sector, as the system ensures safe and tamper-proof

storage for sensitive medical records. Leveraging blockchain's cryptographic algorithms, the solution provides an added layer of security, making it resistant to unauthorized access or tampering.





ABSTRACT

The implementation revolves around utilizing a proof-of-work concept with a focus on achieving the desired difficulty level using a randomly generated nonce. The presentation will spotlight the significance of the proof-of-work algorithm as the consensus mechanism, requiring miners to solve cryptographic puzzles before appending blocks to the chain. A critical analysis of two proof-of-work algorithms, focusing on nonce generation methodologies.

Throughout the presentation, the focus remains on the implemented on-chain blockchain storing comprehensive file data within blocks. Insights from the proof-of-work algorithm analysis contribute to understanding the intricate balance between randomness, security, and computational efficiency in decentralized file storage systems.



INTRODUCTION

- In today's digital landscape, data security and integrity stand as paramount concerns, particularly in sectors handling sensitive information like healthcare. The advent of blockchain technology has sparked innovative solutions to address these challenges, offering immutable and secure data storage systems.
- This presentation delves into the realm of decentralized file storage leveraging blockchain, focusing on its implementation and the critical analysis of proof-of-work methodologies. Through this exploration, we aim to elucidate the intricacies of a web-based application designed to facilitate secure file storage within a blockchain network.
- The fundamental premise lies in ensuring robust data integrity, particularly in contexts such as healthcare, where sensitive medical records demand the highest standards of confidentiality and tamper resistance. By harnessing the cryptographic capabilities of blockchain, our solution endeavors to provide an impregnable fortress for the safekeeping of vital patient information.
- Throughout this presentation, we'll navigate the core principles, implementation intricacies, and comparative analysis of proof-of-work algorithms, shedding light on their impact on system functionality, security, and efficiency. Join us on this journey as we explore the convergence of blockchain technology and decentralized file storage, unraveling the potential it holds in ensuring secure data ecosystems.

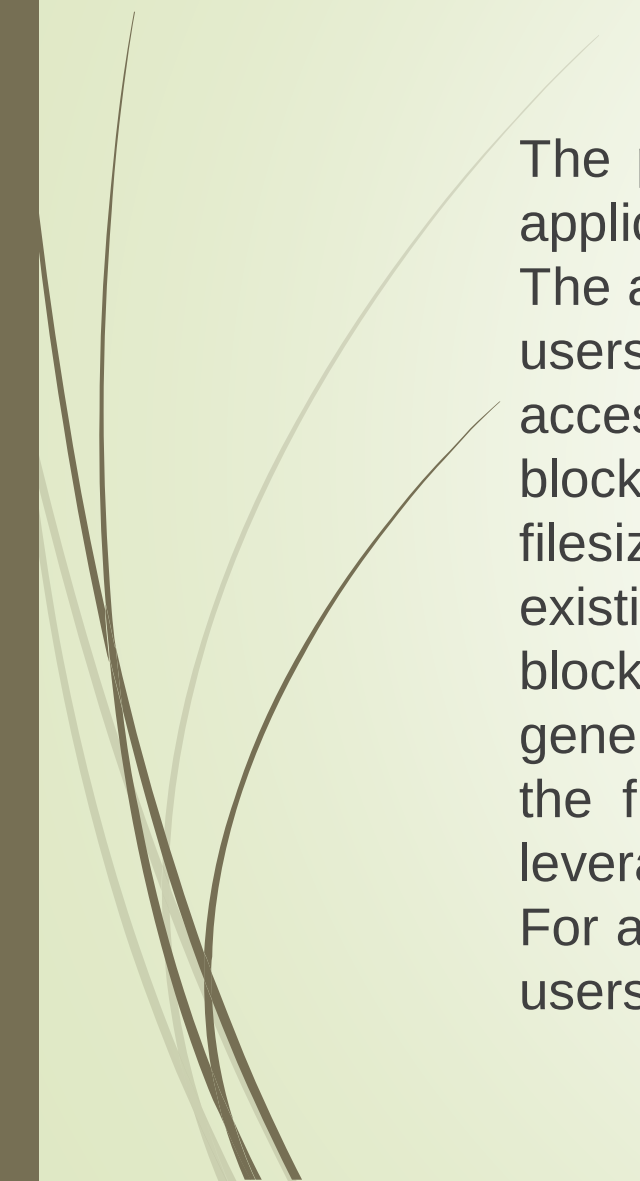
EXISTING SYSTEMS

This already exists as paper work and the implementing the localised shared storage for storing medical record is to be done in this project





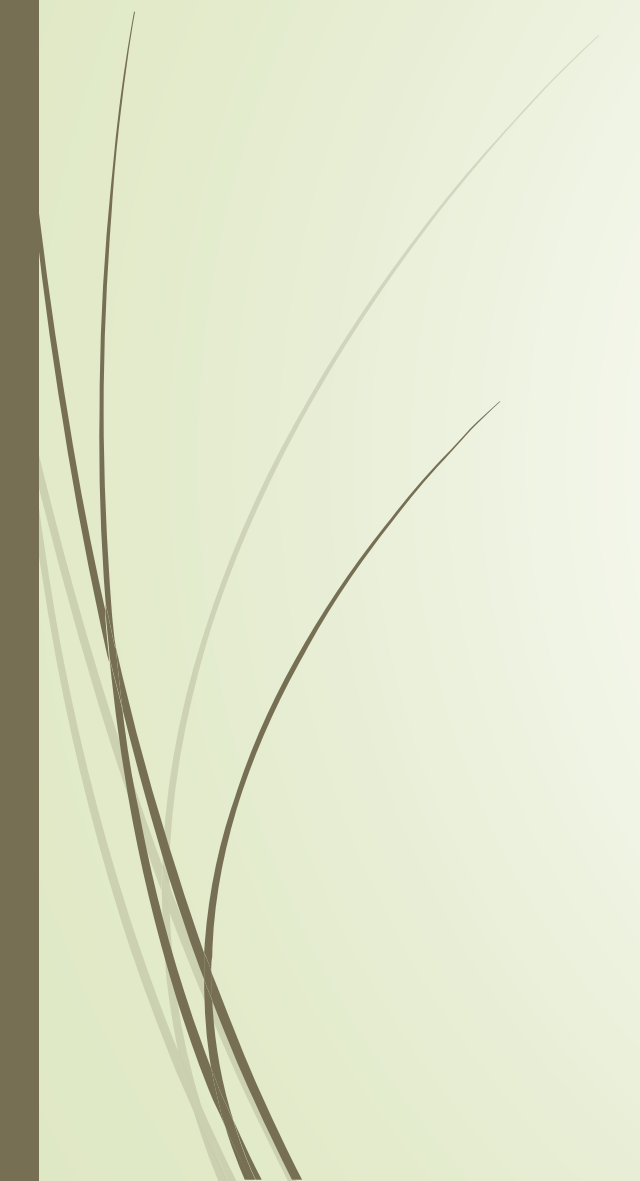
PROBLEMS DEFINED



The problem addressed in this project centres on the creation of a web-based application designed for decentralized file storage using blockchain technology. The application allows users to upload files of any type and size, one at a time. All users, including the uploader and other peers in the network, can download and access these files. To ensure security and immutability, the system employs a blockchain structure. When a user uploads a file, its details, including username, filesize, and file data, are stored in a block. These blocks are then appended to the existing blockchain, making it virtually impossible to edit or delete the files or blocks. The implementation incorporates a proof of work concept with randomly generated nonces to achieve a specified difficulty level, enhancing the security of the file storage process. The underlying motivation for using blockchain is to leverage its ability to safeguard files against unauthorized modification or deletion. For a detailed explanation and demonstration, the project provides a demo link for users to explore further.



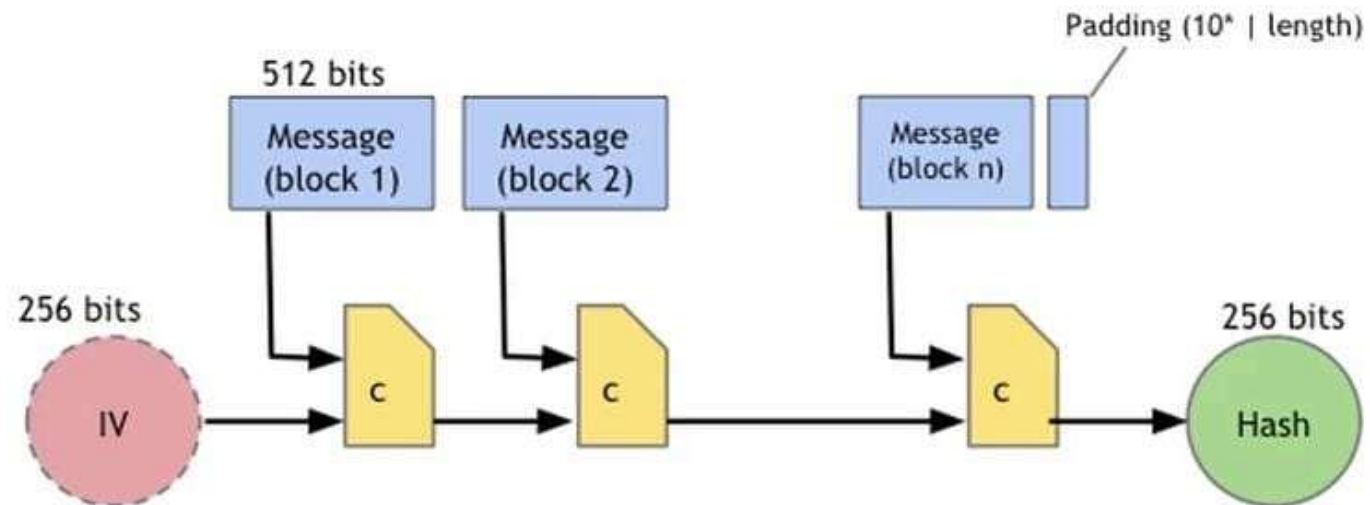
PROPOSED SYSTEM

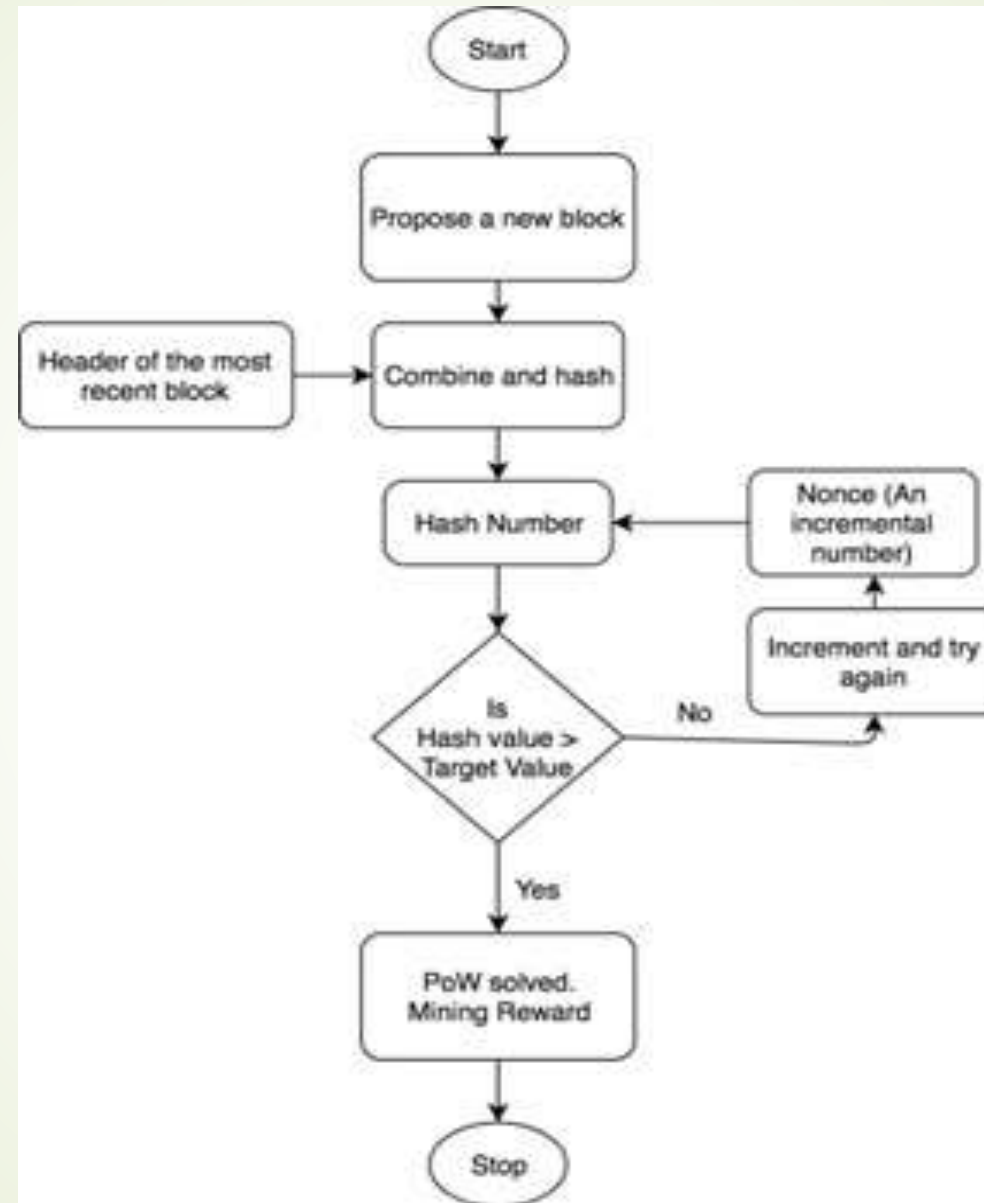


In The proposed system, we have used SHA256 cryptographic algorithm to secure the block. As a consensus algorithm, proof of work is implemented, which requires miners to solve any cryptographic puzzle before they get to announce new block on chain. In our application, we require to find a hash value that starts with three 0's as part of a puzzle.

SYSTEM ARCHITECTURE

SHA-256 hash function







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